

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. (EC) Examination

May 2014

EC 310.01 Optical Communication

Date: 12.05.2014, Monday

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Discuss the key elements of optical fiber communication system. [04]
(b) Compare step index and graded index fibers. [03]
- Q - 2 (a) What do you mean by degradation of signal in optical fiber? Describe bending losses in detail. [05]
(b) Explain Fabry Perot resonator cavity for LASER diode with neat figure. [05]
(c) A 30 Km long optical fiber has an attenuation of 0.4dB/km at 1310 nm. Find out the output optical power if 200 μ W of optical power is launched into the fiber [04]

OR

- Q - 2 (a) Describe with a neat and labeled sketch the construction of edge emitting LED [05]
(b) A double heterojunction InGaAsP LED emitting at a peak wavelength of 1310 nm has radiative and non-radiative recombination times of 30 ns and 100 ns respectively. The drive current is 40mA. Calculate internally power generated to the LED. [05]
(c) What do you mean by dispersion? Describe Chromatic dispersion in brief. [04]
- Q - 3 (a) Discuss the most common type of mechanical misalignment occurring between two joined fibers and its effect. [06]
(b) Justify the following: [06]
(i) Numerical aperture is also a deciding factor for number of modes.
(ii) The addition of dopants of silica causes variation in refractive indices to the core and cladding.
(c) Define (i) Fiber beat length. (ii) Birefringence [02]

- Q - 3 (a) Derive the expression for the power launched from the surface emitting LED source to step index and graded index fiber for radius of the source greater than the core radius. [06]
- (b) Enlist types of fiber fabrication process and explain OVPO process in detail with necessary diagram. [06]
- (c) Find the core radius necessary for single mode operation at 1320nm of a step index fiber with $n_1=1.480$ & $n_2=1.478$. What is the NA of this fiber? [02]

SECTION - II

- Q - 4 (a) Define following terms with reference to Photo detector:
(i) Responsivity (ii) Quantum efficiency (iii) Photo current [03]
- (b) What is switching? Give the Difference between Packet switching and Brust switching. [04]
- Q - 5 (a) Explain principle of operation, the structure and the characteristics of the pin diode. [05]
- (b) Draw and explain schematic diagram of a typical optical receiver. [05]
- (c) What are the different types of pre amplifiers? Explain any one of them in brief [04]

OR

- Q - 5 (a) Compare silicon RAPD with PIN photo detector with respect to the structure, principle of operation and the performance. State the advantages of RAPD [05]
- (b) Briefly discuss digital signal transmission in optical cable. [05]
- (c) A photodiode has a quantum efficiency of 65% when photons of energy 1.5×10^{-19} J are incident upon it. [04]
- (i) At what wavelength is photo diode operating?
- (ii) Calculate incident power required to obtain a photocurrent of $2.5\mu A$ when the photodiode is operating as described above.
- Q - 6 (a) Explain SONET frame structure Give the difference between SONET and SDH. [05]
- (b) Discuss $N \times N$ star coupler with necessary equation. [05]
- (c) Which factors affect the Rise Time Budget? In One optical communication link have the photo diode which required receiver input signal of -42dbm and optical source that can transmit -13dbm average optical power level in to the fiber fly head with $50\mu m$ core diameter. Assume that 1db loss occur when the fiber fly head is connected to the cable and another 1db connector loss. Link also includes the 6db system margin. Find the possible transmission distant with $\alpha f = 3.5db/km$ [04]

OR

- Q - 6 (a) Define Tunable sources. Enlist types of it and Discuss relationship between tuning range, channel spacing and spectral width. [05]
- (b) Explain the Subcarrier Multiplexing. [05]
- (c) Define EDFA. What are the key system features of WDM? [04]
